

Data for Policy Analysis and Management

CLASS 5: DERIVED VARIABLES
AND LINEAR REGRESSION

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Agenda

1. Derived Variables
2. Linear Regression
3. OLS:
 - Coefficients
 - Assumptions
4. Reading Regression Outputs
5. Model Fit, Statistical Significance and Statistical Power
6. Small Group Activity
- 7. Lab: Linear Regression

Derived Variables

What is a Derived Variable?

- A new variable created by combining, recoding, or transforming existing variables.
- They reorganize or summarize existing data but do not add new information.

Why Create Derived Variables?

- Simplify analyses and interpretation.
- Combine related information into a single measure.
- Create variables that better answer a research question.
- Examples: race/ethnicity in AFCARS; any disability (1=Yes if ≥ 1 disability); number of disabilities per child.





Linear Regression

Used Across Disciplines

- Linear regression is one of the most widely used statistical methods in economics, public policy, education, public health, and the social sciences.

What Researchers Use It For

- Describe relationships between variables.
- Predict outcomes from one or more predictors.
- Estimate how much an outcome changes per unit change in a predictor.
- Control for other variables that may influence the outcome.
- Test hypotheses about relationships between variables.
- Estimate causal effects when design and assumptions support it.

Does education increase earnings?

Does teacher experience improve student achievement?

Does participation in a workforce grant increase provider retention?

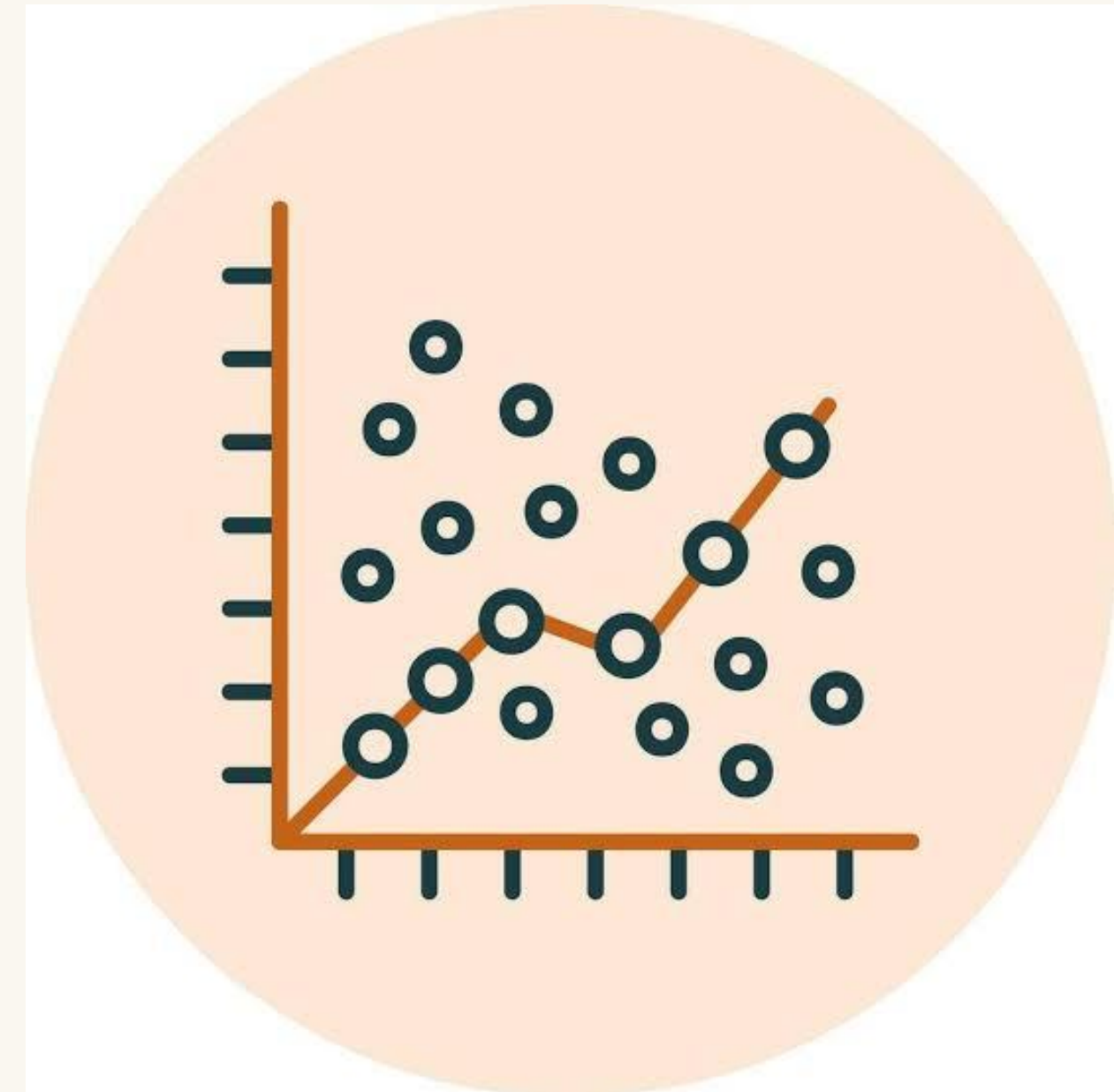
What Does a Linear Regression Actually Do?

Core Function

- Estimates the best-fitting line that summarizes the relationship between an explanatory variable (X) and an outcome variable (Y).

Unlike Correlation, Regression:

- Estimates an equation describing the relationship.
- Measures both direction and magnitude of the relationship.
- Allows prediction of outcomes for given values of X.
- Can include multiple explanatory variables simultaneously.



Linear Regression

The Equation

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

Where:

Y = outcome (dependent variable)

X = predictor (independent variable)

β_0 = intercept

β_1 = slope

ε = error term

Interpreting the Line

- The regression line shows the **average expected value of Y for each value of X** .
- Individuals with the same X can have different outcomes due to the **error term, which captures unobserved factors**.
- The slope β_1 **measures the direction and magnitude of the relationship between X and Y** .
- The intercept β_0 **is the expected value of Y when X equals zero**.

Ordinary Least Squares (OLS)

How OLS Works

- Regression estimates are obtained via OLS.
- OLS finds the best-fitting line by minimizing the sum of squared residuals (the squared differences between observed and predicted values) to reduce overall prediction error.

Why Square the Residuals?

- Prevents positive and negative errors from canceling each other out.
- Gives greater weight to larger errors, penalizing big mistakes more.
- Produces a unique, mathematically optimal best-fitting line.
- Ensures a single, consistent solution for any dataset.



Predicted Values, Residuals, and Errors

Observed, Predicted & Residual

- After regression estimation, each observation has three values:
- Observed value (Y): the actual outcome recorded in the data
- Predicted value ($\hat{Y}$): the value the model estimates based on X
- Residual = $Y - \hat{Y}$: the difference between observed and predicted
- Residuals represent how far each data point falls from the fitted regression line.

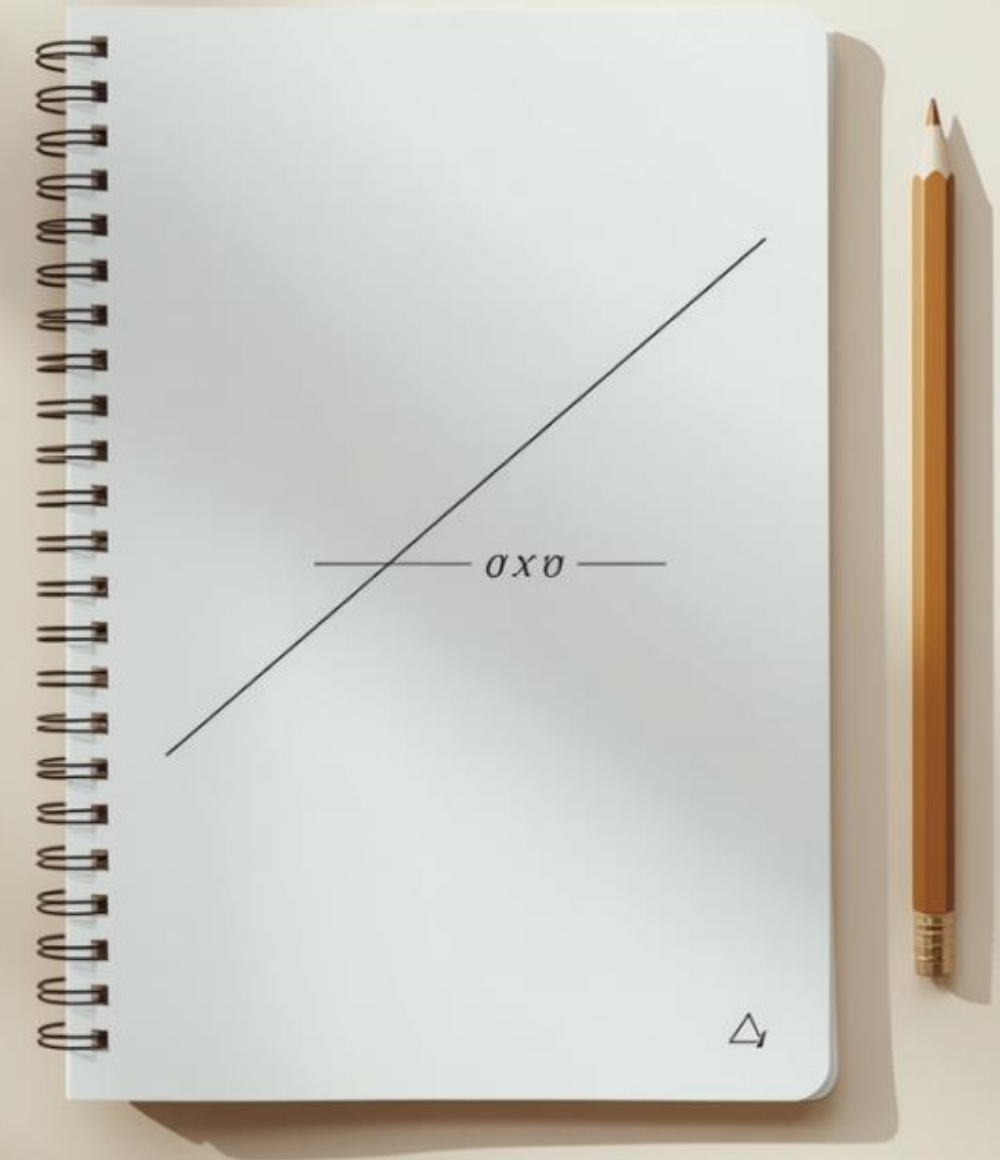
Residuals VS. Error term

- The error term (ε) represents **unobserved factors** that affect the outcome but are not included in the model.
- Residuals are the **sample estimates of these unobserved errors**.

Interpreting OLS Regression Coefficients

$$\text{Wage} = 12.5 + 0.80 * \text{Years of Education}$$

- **Intercept (12.5):** The expected hourly wage when education = 0 years is \$12.5.
- **Slope (0.80):** Each additional year of education is associated with an increase of \$0.80 in hourly wage, on average, holding all other variables constant.
- Remember, *regression estimates average relationships across observations, not exact individual predictions.* The same equation may not perfectly describe any single person's outcome.

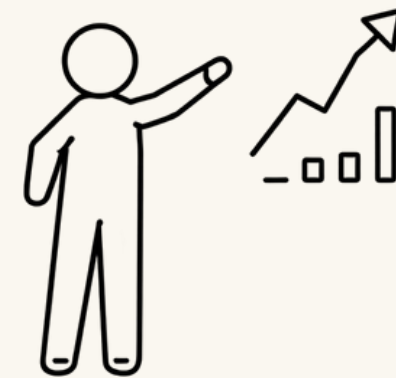


OLS Assumptions

Regression provides unbiased and reliable estimates only when several assumptions are approximately satisfied.

These assumptions ensure:

- **unbiased** coefficient estimates.
- valid standard errors.
- trustworthy confidence intervals.
- accurate hypothesis tests.
- reliable predictions.



A biased estimate is an estimate that consistently overestimates or underestimates the true value

OLS Assumptions

1. Linearity

Relationship between X and Y should be approximately linear.

1. Exogeneity

Explanatory variables must not correlate with the error term. Violations cause omitted variable bias, distorting coefficient estimates.

1. Homoscedasticity

The variability of the residuals (prediction errors) is approximately constant across all values of the independent variable

1. Independence

Each observation is independent from each other. Violations (clustering, repeated measures) affect standard error accuracy.

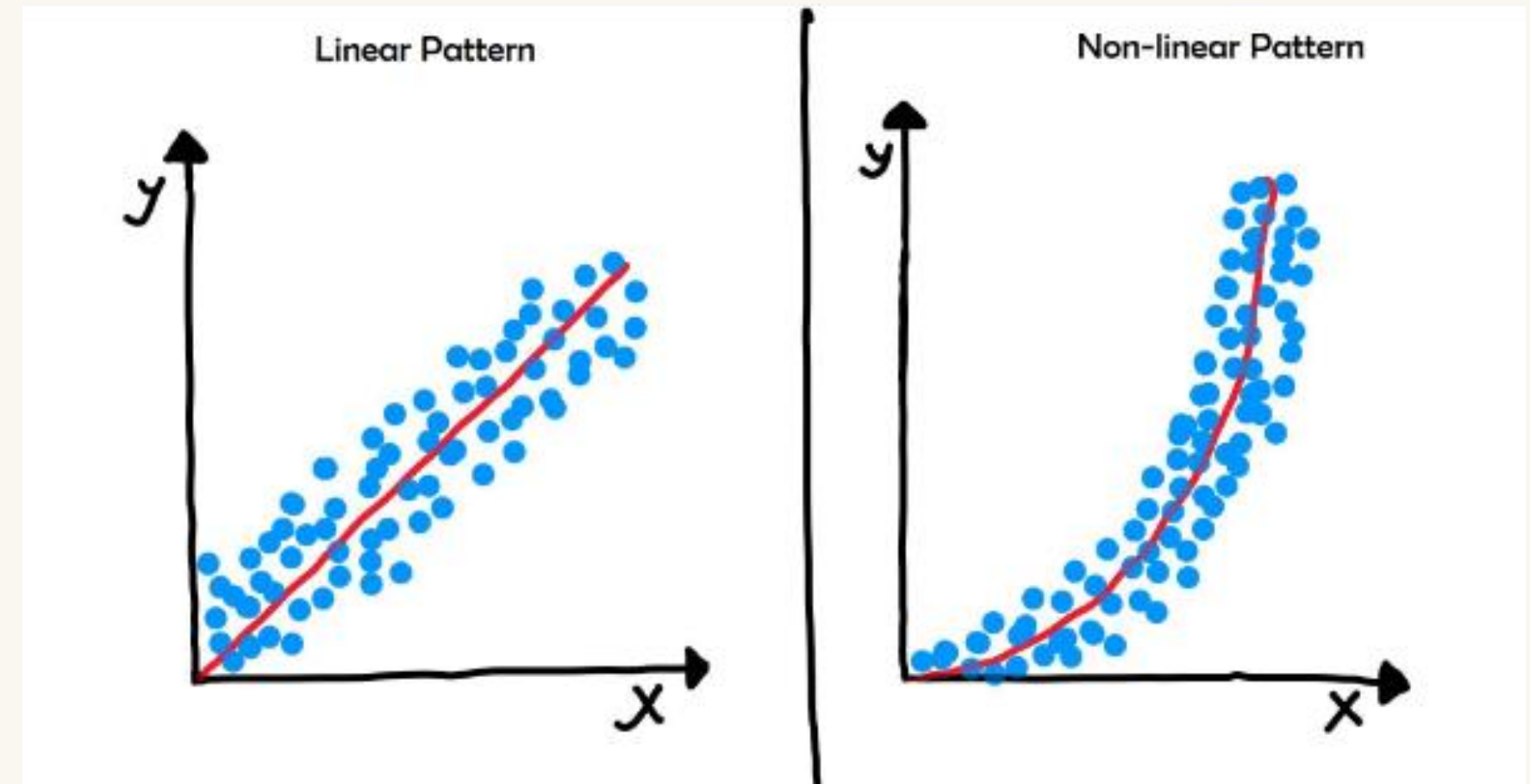
1. Normality

Residuals should be approximately normally distributed.

Assumption 1: Linearity

What It Requires

- The relationship between X and Y should be approximately linear.
- This means that as X increases, Y changes at a roughly constant rate (forming a straight-line pattern in the data).
- If the true relationship is curved but we fit a straight line, our predictions and coefficient estimates may be misleading.



Diagnostic Tools

- Scatterplots: Look for a roughly linear pattern. A curved or non-linear pattern suggests the linearity assumption may be violated.
- Residual plots: Plot residuals against fitted values or X. Residuals should be randomly scattered around zero with no clear pattern. A curved pattern in the residuals signals non-linearity.
- If non-linearity is detected, consider transforming variables (e.g., log transformation) or adding polynomial terms.

Assumption 2: Independence

TEAM A

What It Means

- Each observation must provide independent information, meaning the value of one observation should not influence or be related to another.
- Example of independence met:
 - Different individuals selected via random sampling ✓
- This ensures that each data point contributes unique, non-redundant information to the analysis.

When Independence Fails

Common violations of independence:

- Repeated observations on the same person over time.
- Clustered observations (e.g., students within schools).

Consequences:

- Violations primarily affect standard error accuracy, making hypothesis tests and confidence intervals untrustworthy.
- Coefficients may remain unbiased.

Assumption 3: Exogeneity

What It Requires

- Explanatory variables (X) should not be correlated with the error term (ε).
- If X and ε are correlated, regression coefficients will be biased and inconsistent (they will not accurately reflect the true relationship between X and Y).
- This is one of the **most critical assumptions** for obtaining valid causal estimates.

Example: Estimating the effect of education on earnings while omitting work experience

- Work experience is related to both education level and wages.
- Omitting it places its effect into the error term.
- Since experience correlates with education, the exogeneity assumption is violated.
- The education coefficient absorbs part of the experience effect, biasing the estimate.
- Remedy: *Include relevant control variables to reduce omitted variable bias.*

Assumption 4: Homoscedasticity

What Is Homoscedasticity?

- Homoscedasticity means **the spread of residuals is roughly constant** across all values of X .
- In other words, the variance of the error term does not change systematically as the predictor changes.

What Is Heteroscedasticity?

- Heteroscedasticity occurs when residual variance changes systematically across values of X .
- For example, residuals that fan out or compress as X increases. This is a violation of the constant variance assumption.

Effects on Estimates

When heteroscedasticity is present:

- Coefficient estimates remain unbiased
- Standard errors become unreliable
- Hypothesis tests are less trustworthy
- Confidence intervals may be too wide or too narrow

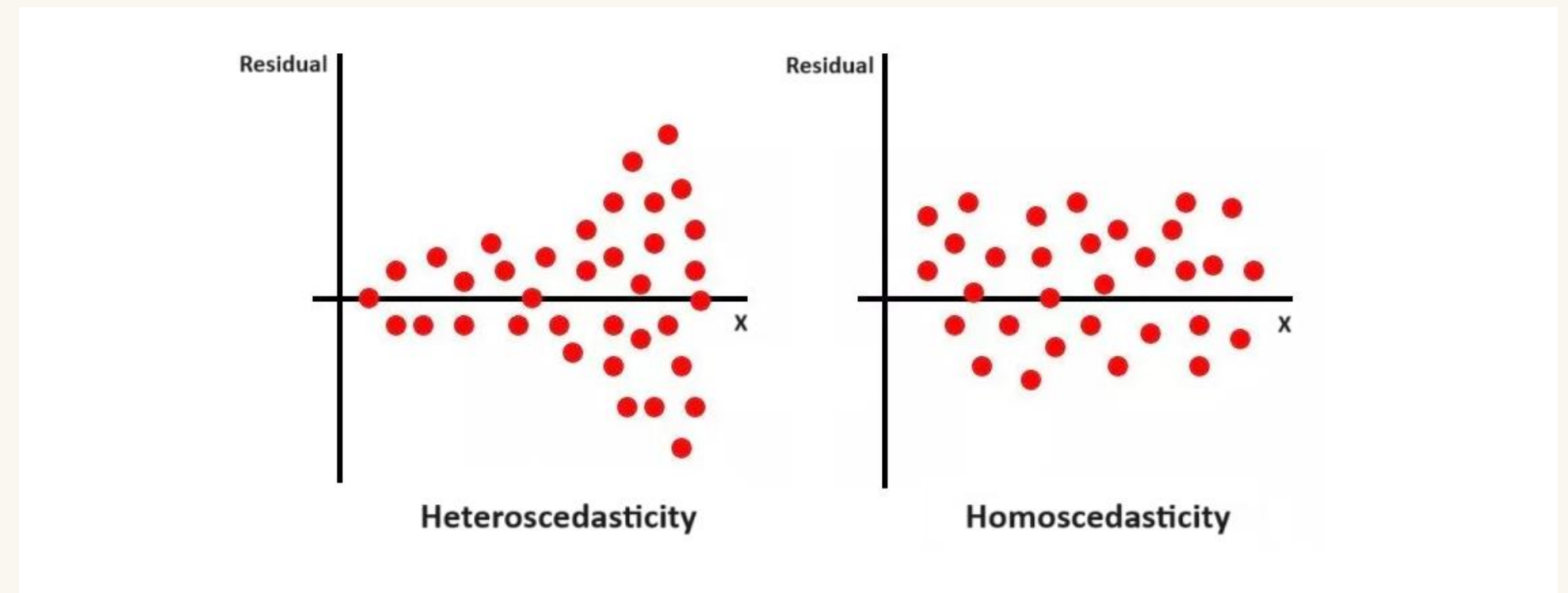
How to Detect & Address It

- Detection: Examine residual plots --> look for patterns in spread as X increases. Formal tests can also be used.
- Remedies: Use robust standard errors, transform the outcome variable, or apply weighted least squares.

Assumption 4: Homoscedasticity

Example:

- Suppose you're predicting hourly wages based on years of experience.
- If your prediction errors are usually around $\pm\$2$ regardless of whether someone has 1 year or 20 years of experience, that's homoscedasticity.
- If the errors are only $\pm\$1$ for people with little experience but $\pm\$10$ for highly experienced workers, that's heteroscedasticity.



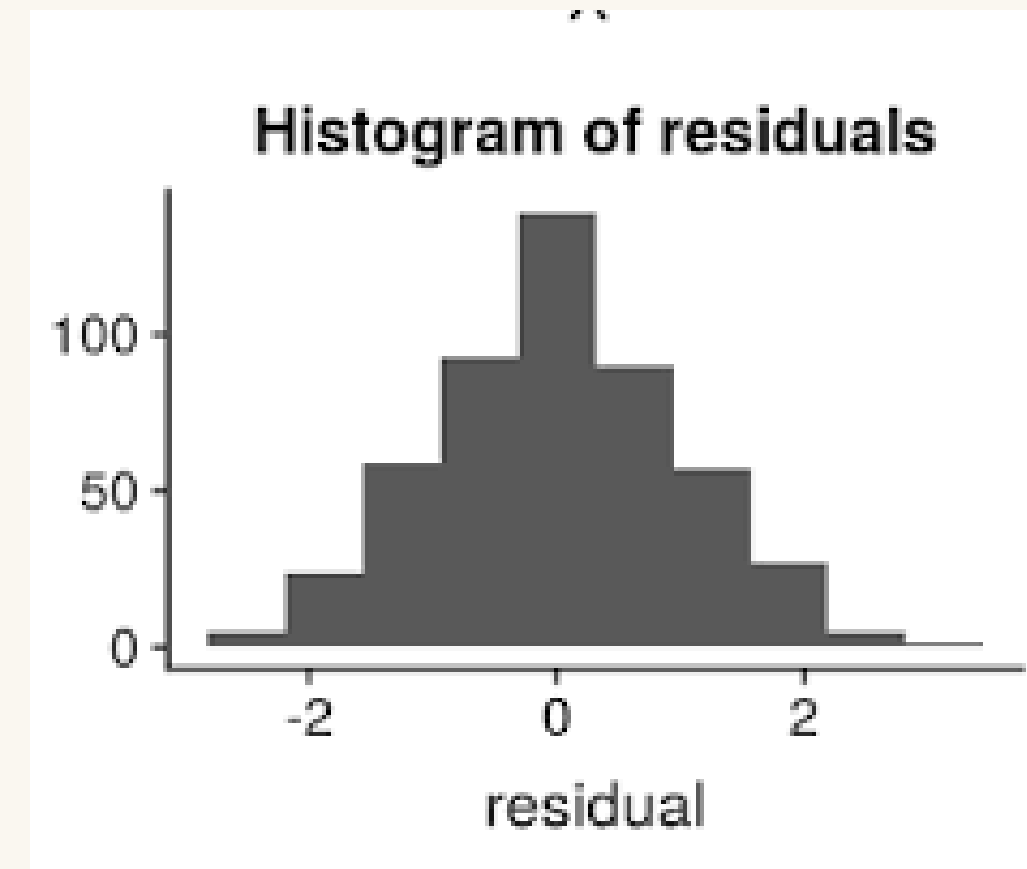
Assumption 5: Normality

What It Requires

- Residuals should be approximately normally distributed around zero.
- This means the errors from the regression model follow a bell-shaped distribution, with most errors near zero and fewer large errors.

How to Evaluate

- Use a histogram of residuals to check for a bell-shaped distribution, or a Q-Q plot where points should fall along a diagonal line.
- Significant departures suggest non-normality.



Why It Matters

- Normality is important for the validity of confidence intervals, hypothesis tests, and p-values.
- Without it, these inferential tools may not be reliable, especially in small samples.

In large samples ($n \geq 30$), the Central Limit Theorem reduces the importance of this assumption.

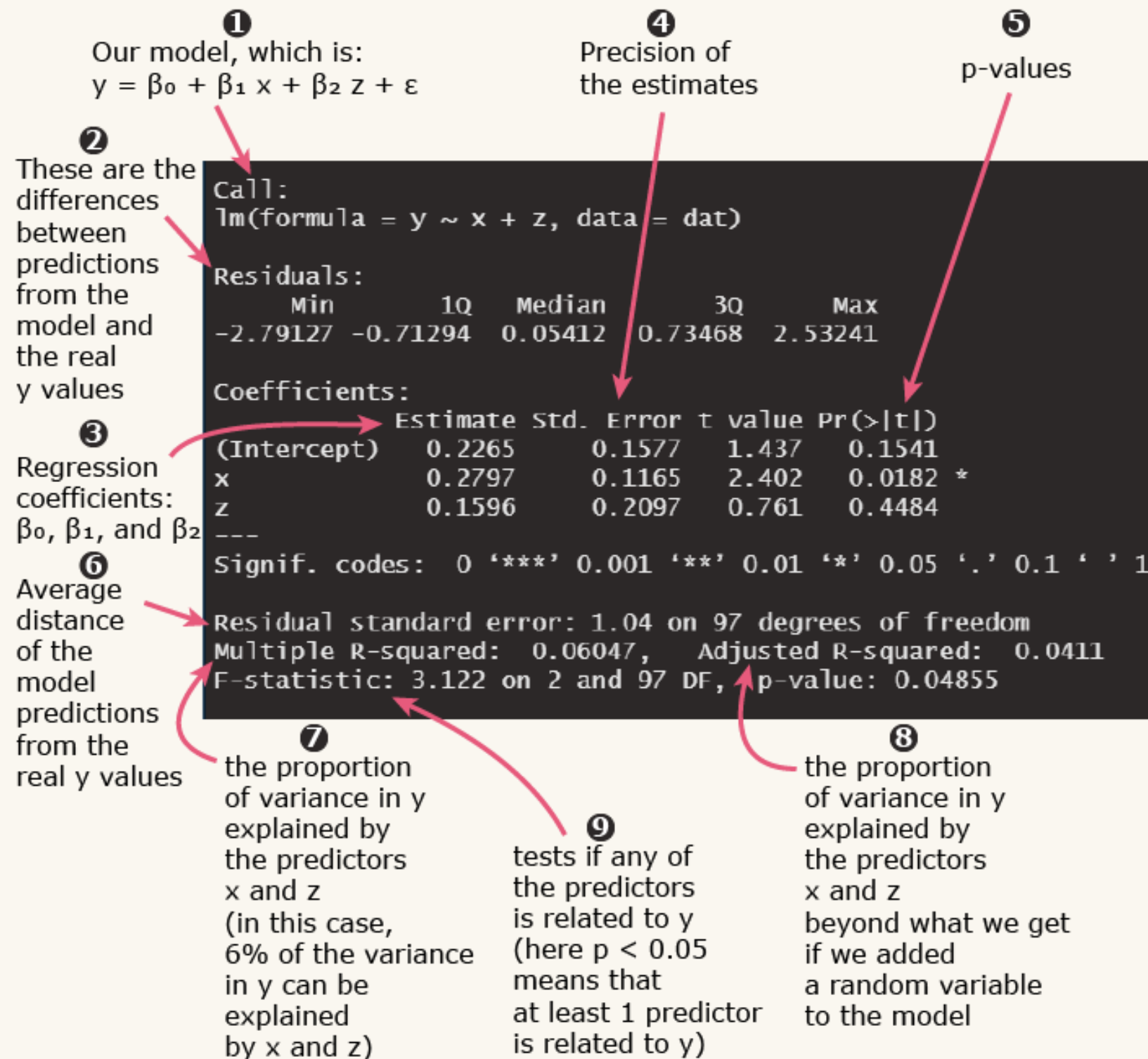
Reading Regression Output

What Regression Output Reports

- Typical regression output includes: coefficient estimates, standard errors, t statistics, p-values, confidence intervals, R^2 and adjusted R^2 , and number of observations.

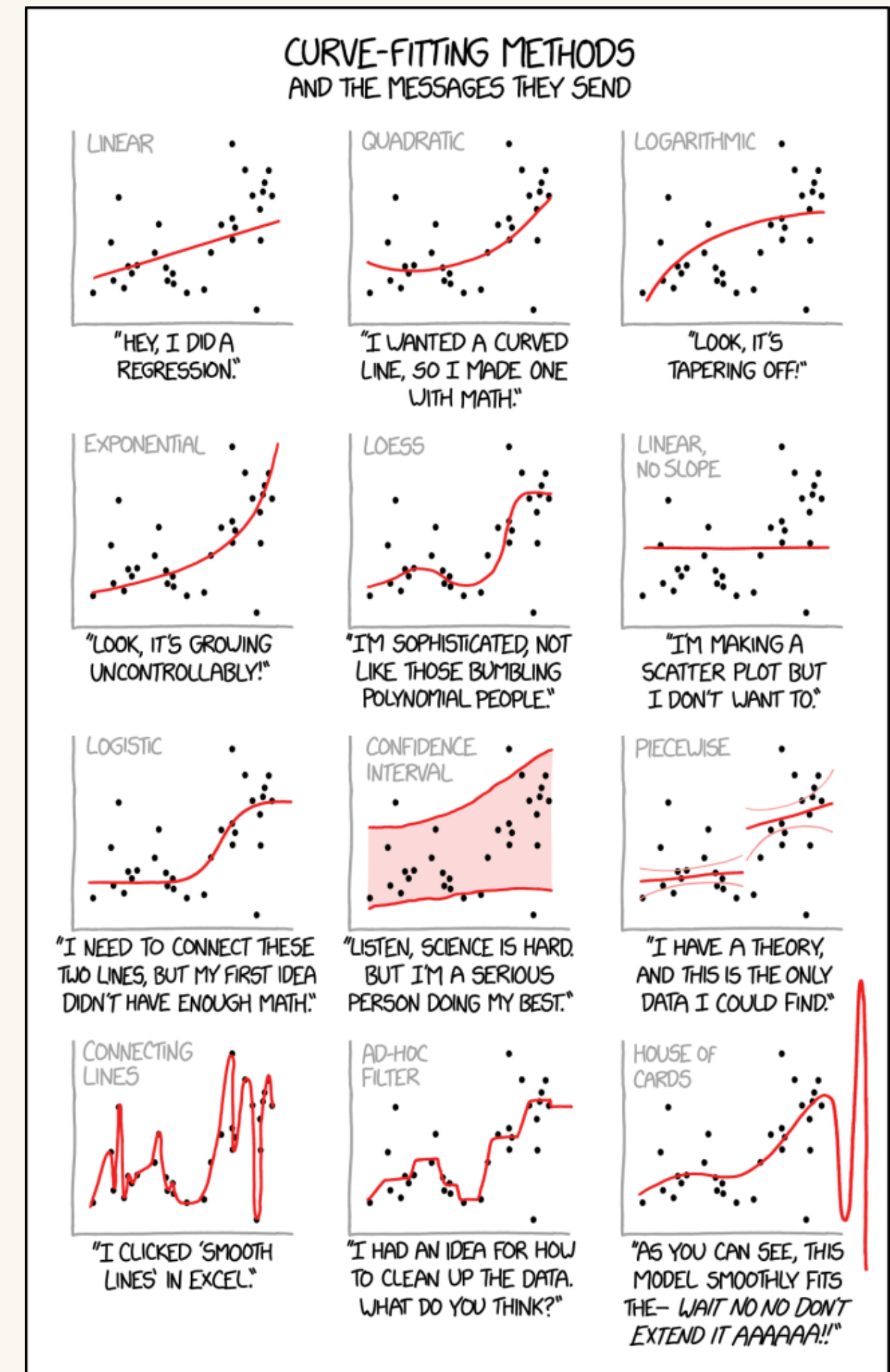
What Each Statistic Tells You

- Coefficient estimate: strength and direction of the relationship.
- Standard error: precision of the coefficient estimate.
- t statistic: coefficient divided by its standard error.
- p-value: probability of result if null hypothesis is true.
- Confidence interval: plausible range for the true coefficient.
- R^2 : proportion of outcome variability explained by the model.
- Adjusted R^2 : R^2 penalized for additional predictors.
- N (observations): sample size used in the estimation.



Understanding Model Fit (R^2)

- R^2 measures the **proportion of variability in the outcome that is explained by the model's predictors**.
- In OLS, $0 \leq R^2 \leq 1$
- Example: $R^2 = 0.70$ means that 70% of the variability in the outcome is explained by the predictors included in the model.
- Key notes:
 - Higher R^2 does not always mean a better model.
 - A low R^2 can be acceptable when the goal is estimating relationships rather than prediction.
 - Focus on your research question rather than maximizing R^2 .
 - Adjusted R^2 penalizes for adding unnecessary predictors.



Statistical Significance

Testing the Null Hypothesis

- For each coefficient, regression tests the null hypothesis $H_0: \beta = 0$, meaning the predictor has no relationship with the outcome.
- A small p-value indicates the observed relationship is unlikely due to chance, suggesting the predictor is associated with the outcome.
- The t-statistic measures how many standard errors the estimate is from zero.

Reminders

- **Statistical significance does not equal practical importance.** A statistically significant effect can be very small and may have little real-world meaning.
- An **non significant result does not prove there is no relationship**, it may simply reflect limited sample size or low precision.
- Always interpret significance alongside effect size and confidence intervals to draw meaningful conclusions.

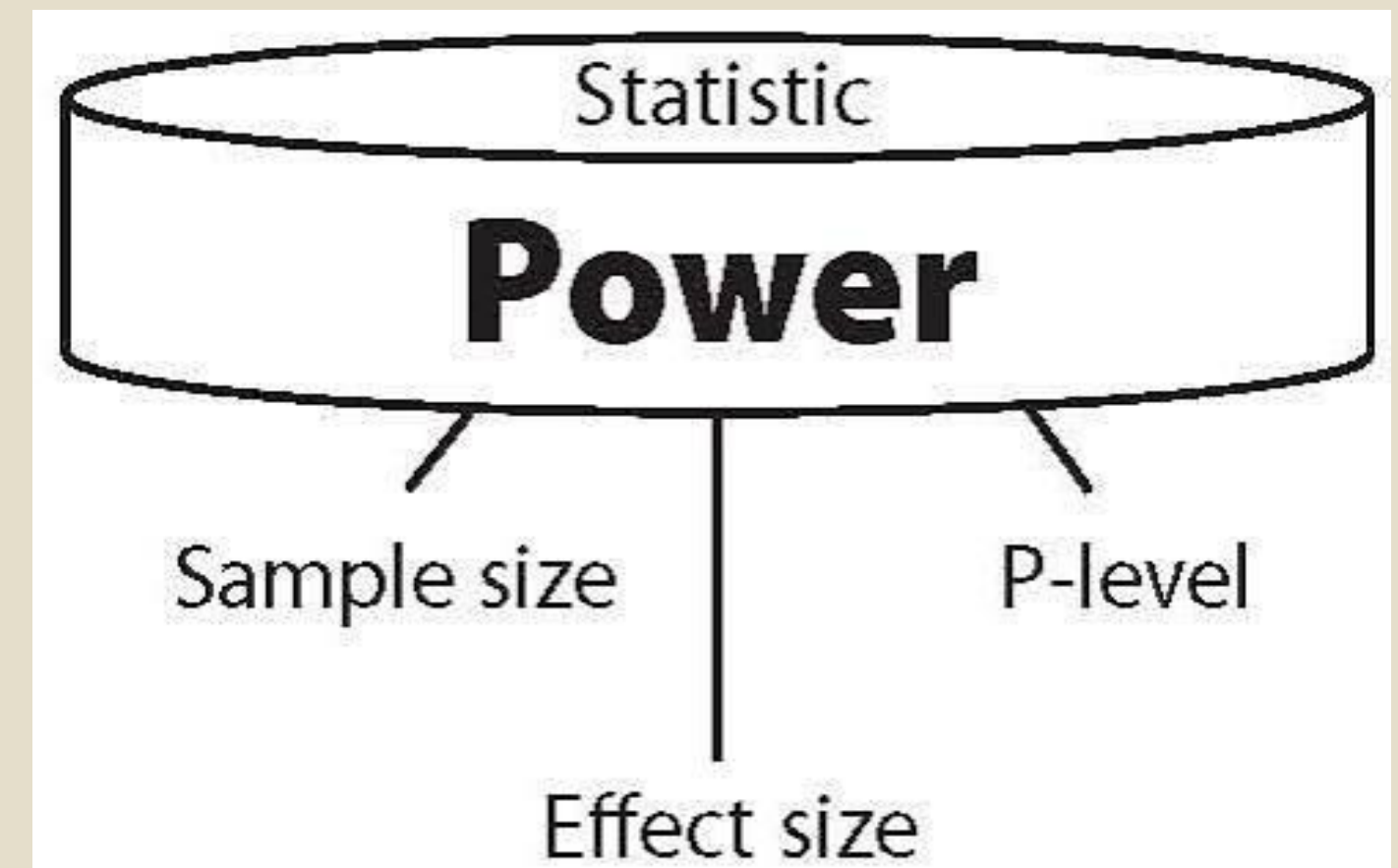
Statistical Power

What It Is

- Statistical power is the ability of a study to find a real relationship or difference instead of missing it.

Why It Matters

- Larger samples increase statistical power, making it more likely that we will detect real effects and less likely that we will incorrectly conclude there is no relationship when one actually exists.



Key Takeaways

$$\hat{\beta}_1 = \frac{\beta_1 \sum (x_i^2 - \bar{x} x_i) + \frac{\sum x_i u_i}{\sum x_i^2}}{\sum (x_i - \bar{x})^2}$$

$$\hat{\beta}_1 = \beta_1 \frac{(\sum x_i^2 - \bar{x} \sum x_i)}{\sum x_i^2 - n \bar{x}^2} + \frac{\sum x_i u_i}{\sum x_i^2}, \text{ let } K_i = \frac{x_i}{\sum x_i^2}$$

$$\hat{\beta}_1 = \beta_1 \frac{(\sum x_i^2 - n \bar{x}^2)}{(\sum x_i^2 - n \bar{x}^2)} + \sum K_i u_i$$

$$\hat{\beta}_1 = \beta_1 + \sum K_i u_i$$

$$\text{Now, } E(\hat{\beta}_1) = E(\beta_1) + \sum E(K_i u_i)$$

$\beta_1 \rightarrow \text{Population Parameter} \rightarrow \text{Constant}$

$$\Rightarrow E(\hat{\beta}_1) = \beta_1 + \left\{ E(K_1 u_1) + E(K_2 u_2) + \dots + E(K_n u_n) \right\}$$

$$= \beta_1 + \left\{ K_1 E(u_1) + K_2 E(u_2) + \dots + K_n E(u_n) \right\}$$

[$\because$ for K_i , there are many u_i]

$$\text{Also, } E(u_i) = 0$$

$$\Rightarrow \boxed{E(\hat{\beta}_1) = \beta_1} \Rightarrow \hat{\beta}_1 \text{ is the } \underline{\text{unbiased}} \text{ estimator of } \beta_1.$$

- Linear regression estimates the **average** relationship between variables.
- OLS chooses the line that minimizes prediction errors.
- Regression coefficients quantify expected changes in the outcome.
- **Reliable inference depends on several assumptions**, particularly linearity, independence, exogeneity, homoscedasticity, and normality.
- Statistical significance should always be interpreted alongside effect size, confidence intervals, and substantive importance.
- Regression is a powerful tool for explanation and prediction, but **causal interpretation requires an appropriate research design and valid assumptions**.

Small Group Activity

Work in groups of 4, discuss the following questions and be ready to share your reasoning.

Pilkauskas et al. (2014) examine the prevalence of doubling up among families with young children in large U.S. cities and estimate its economic value. Using information from this paper, respond to the following questions:

- How do the authors define "doubling up"? How is this concept measured in the study?
- Would you use an OLS regression model to estimate the effect of doubling up on family income? Why or why not? Be sure to discuss whether the assumptions of OLS are likely to hold in this context.
- Suppose you wanted to examine the factors associated with doubling up using a regression model. What variables would you include as independent (predictor) variables? What important factors would be difficult to measure and would likely be captured in the model's error term?

Next Class

- Linear Regression cont.
- Bressane et al. paper

Questions?

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15-minute break